Task A

> Crucial Notice before you read and mark my work
>
> Mac-specific session setup. macOS BSD tools (cut, grep, tr) error out on non-UTF-8 bytes in the property descriptions, producing cut: stdin: Illegal byte sequence and silently truncated output. The fix is export LC_ALL=C, which switches the shell to byte-mode. Note: this only lasts for the current terminal session — closing the terminal resets it. On returning to the assignment, I observed this on Q3 when re-opening Terminal a day later: my counts initially diverged from the previous day's results, and re-applying LC_ALL=C restored them. Due to I finish Q1, Q2 in one day, and Q3 - Q6 in another, only applied LC_ALL=C twice in my code block you can see.

Q1 — sold_time range of the records

Solution1

Approach

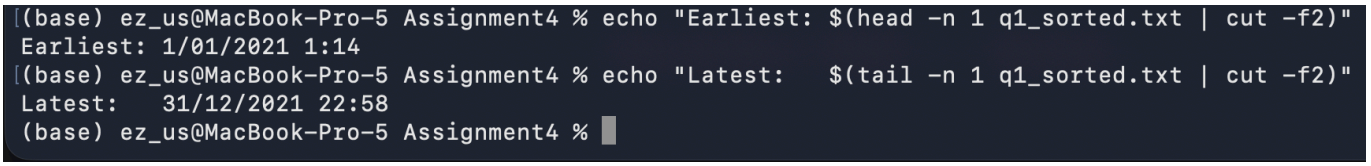
`sold_time` has the form `"D/M/YYYY H:MM"` but is **not** zero-padded format (e.g. `8/11/2021 16:33` sits beside `27/03/2021 22:17`), which leads to wrong `sort`. So convert each value to a sortable form `YYYY-MM-DD HH:MM` before sorting.

Code

```
# Step 1: extract sold_time column, skip header, drop returns,
# and keep only values that look like a date+time

tail -n +2 TaskA_property_victoria.csv | cut -d, -f4 | tr -d '\r' | grep -E
'^[0-9]+/[0-9]+/[0-9]+ [0-9]+:[0-9]+$' > q1_sold_times.txt  # output as a
new file
# Extended Regular Expressions, which allows you to use complex patterns.
# Anwser is 48480 q1_sold_times.txt

# Step 2: convert each "D/M/YYYY H:MM" into "YYYY-MM-DD HH:MM<TAB>original"
# The -F'[ /:]' tells awk to split on space, slash, OR colon, so
# "27/03/2021 22:17" becomes 5 fields: 27, 03, 2021, 22, 17.

awk -F'[ /:]' '{ printf "%04d-%02d-%02d %02d:%02d\t%s\n", $3, $2, $1, $4,
$5, $0 }' q1_sold_times.txt | sort > q1_sorted.txt

# peek: top of sorted list
head -n 2 q1_sorted.txt
# output
# 0420210101 01:00  1/01/2021 8:01
# 0420210101 03:00  1/01/2021 22:03

# peek: bottom of sorted list
tail -n 2 q1_sorted.txt
```

```
# output:
# 0420211231 58:00  31/12/2021 22:58
# 0420211231 58:00  31/12/2021 7:58

# Print the result
echo "Earliest: $(head -n 1 q1_sorted.txt | cut -f2)"
echo "Latest:   $(tail -n 1 q1_sorted.txt | cut -f2)"
```

Screenshot



Answer

The `sold_time` range of the records is **1 January 2021 at 1:14** to **31 December 2021 at 22:58**.

Explaination

- Isolate `sold_time` column.
    - `cut -d, -f4`, cut the 4th column accordding to assignment desctiption.
    - `tr -d 'r'`, removes 'return' character.
    - `grep ...` filter rows with valid time format.
    - output it as a new text file: q1_sold_times.txt, which include 48480 data.
- Sort sold_times.
    - Standardise time format. current date would look like "M/D/YYYY H:S" or "MM/DD/YYYY HH:SS", Which determined by whether have "0" or not on date, and 24-hour time standard. It lead to disorder. Parse date form first, then sort.
    - `awk -F'[ /:]'` splits a string like `"27/03/2021 22:17"` on any of space, slash, or colon — so we get the five fields `27`, `03`, `2021`, `22`, `17`.
    - Then `printf"%04d-%02d-%02d %02d:%02d\t%s\n", $3(year), $2(month), $1(day), $4(hour), $5(minuete), $0(Placeholder: No second here)` rearranges them year-first and prints both the sortable form and the original.
    - `sort` orders the lines alphabetically — but because our key iszero-padded and year-first, alphabetical order IS chronological order.
- Strip off the first and last line to get answers
    - `echo` can be regarded as `print` or `cat`(R)

Solucion 2

While I rerun comment below, it turned an error `cut: stdin: Illegal byte sequence`. During figuing out what it is. I noticed there are an OS mismatch error.

```
tail -n +2 TaskA_property_victoria.csv | cut -d, -f4 | tr -d '\r' | grep -E
'^[0-9]+/[0-9]+/[0-9]+ [0-9]+:[0-9]+$' > q1_sold_times.txt
```

Hence, here is another version for anwers.

Code 2

```
cd ~/Documents/5145/Assignment4/TaskA_shell

# Switch the whole session to byte-mode (fixes the Illegal byte sequence)
export LC_ALL=C

# Confirm — should now print "C"
echo $LC_ALL

# remove the generated text files before
rm -f q1_sold_times.txt q1_sorted.txt q2a_ids.txt

tail -n +2 TaskA_property_victoria.csv | cut -d, -f4 | tr -d '\r' | grep -E
'^[0-9]+/[0-9]+/[0-9]+ [0-9]+:[0-9]+$' > q1_sold_times.txt

wc -l q1_sold_times.txt
# turn to 127699, compare 48480 before.

awk -F'[ /:]' '{ printf "%04d-%02d-%02d %02d:%02d\t%s\n", $3, $2, $1, $4,
$5, $0 }' q1_sold_times.txt | sort > q1_sorted.txt

echo "Earliest: $(head -n 1 q1_sorted.txt | cut -f2)"
echo "Latest:   $(tail -n 1 q1_sorted.txt | cut -f2)"
```

Screenshot

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Earliest: $(head -n 1 q1_sorted.txt | cut -f2)"
Earliest: 01/01/2020 00:23
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Latest:   $(tail -n 1 q1_sorted.txt | cut -f2)"
Latest:   31/12/2021 23:58
```

Answer

The `sold_time` range of the records is **1 January 2020 at 00:23** to **31 December 2021 at 23:58**.

Note: the assignment brief says the file is 2021 transactions, but the data actually starts on 1 January 2020.

Q2 - Preprocess the ID(f1) and sold_time(f4) column.

Q2a - Count lines aren't 6-digit long.

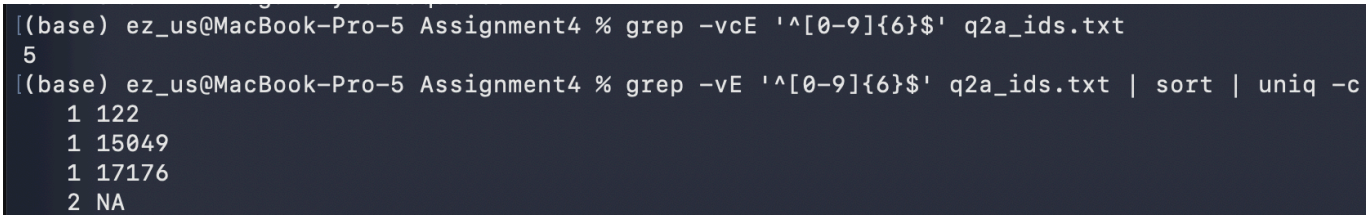
Approach

regex `^[0-9]{6}$` would be valid ID

Code

```
# Step 1: extract ID column, skip header
tail -n +2 TaskA_property_victoria.csv | cut -d, -f1 > q2a_ids.txt

# Step 2: count rows whose ID isn't a 6-digit long nuber
# grep -v inverts the match; -c means 'count'
grep -vcE '^[0-9]{6}$' q2a_ids.txt
```

Screenshot



Answer

5 rows have an `ID` that is not a 6-digit number: 3 rows whose IDs are short numbers(`122`, `15049`, `17176`), and 2 rows where the ID is the literal string `NA`.

Due to mismatch finded on Question one, re-run question 2 would get different anwer as well.

```
# re-run Q2a

tail -n +2 TaskA_property_victoria.csv | cut -d, -f1 > q2a_ids.txt
grep -vcE '^[0-9]{6}$' q2a_ids.txt
# turn to 9, compared to 5 before.

grep -vE  '^[0-9]{6}$' q2a_ids.txt | sort | uniq -c
```

Answer

9 rows have an `ID` that is not a 6-digit number: 5 rows where the ID is the literal string `NA`, and 4 rows whose IDs are short numbers (`122`, `2396`, `15049`, `17176`).

Explanation

- `grep -v` keeps lines that **do not** match the pattern. `-c` return to the number of count.
- `^[0-9]{6}` means 6 digits. `$` limited the output **exactly** equal to 6.
- `sort | uniq -c` arrange **unique count** value from small to large.

Q2b — Drop bad-ID rows and strip the time from sold_time

Approach

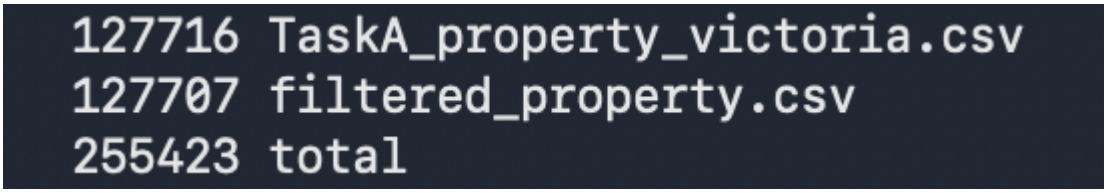
Two transformations:

1. Keep the header plus rows valid ID, which matches `^[0-9]{6}$`.
2. In each kept row, remove the `HH:MM` substring from column 4.

Code

```
# Step 1: extrack a valid ID. Keep header (NR==1) plus rows with 6-digit
long form.
awk -F, 'NR==1 || $1 ~ /^[0-9]{6}$/' TaskA_property_victoria.csv >
q2b_valid_ids.csv

wc -l q2b_valid_ids.csv  # check

# output: 1 header + 127706 data rows = 127707

awk -F, 'BEGIN { OFS = "," } NR == 1 { print; next } { sub(/ [0-9]+:[0-
9]+/, "", $4); print }' q2b_valid_ids.csv > filtered_property.csv

wc -l  # check
# output: 1 header + 127706 data rows = 127707. No miss.

# Verify:
original 127716 - 9 bad rows = 127707 lines (header + 127706 data rows)
wc -l TaskA_property_victoria.csv filtered_property.csv
```

Screenshot



Explanation

- `awk -F, 'NR==1 || $1 ~ /^[0-9]{6}$/'` is the entire pattern. `NR==1` is true on the header row. `$1 ~ /^[0-9]{6}$/` is true on rows where the ID is exactly 6 digits. `||` means "or"

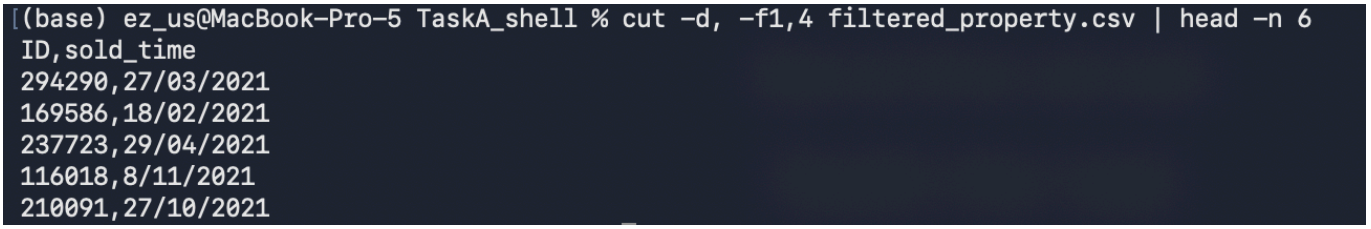
- `sub(/ [0-9]+:[0-9]+/, "", $4)` finds the first occurrence of " H:MM" or " HH:MM" inside column 4 and replaces it with empty. Because we assign to a field, awk rebuilds the line `OFS = ","`: Since the input and output separators are both comma, every other field comes back byte-identical, including rows `description` contained unescaped commas.
- 127716 − 9 = 127707 — counts match Q2a.

Q2c — Display first 5 lines of ID and sold_time

Code

```
cut -d, -f1,4 filtered_property.csv | head -n 6
```

Screenshot



Explanation

- `cut -d, -f1,4` picks columns 1 and 4 (`ID` and `sold_time`); they're safe with `cut` because no commas appear inside the first four columns.
- `head -n 6` 1 header + the first 5 data rows

Q3 — First and last mention of "Mount Dandenong" in `address`

Approach

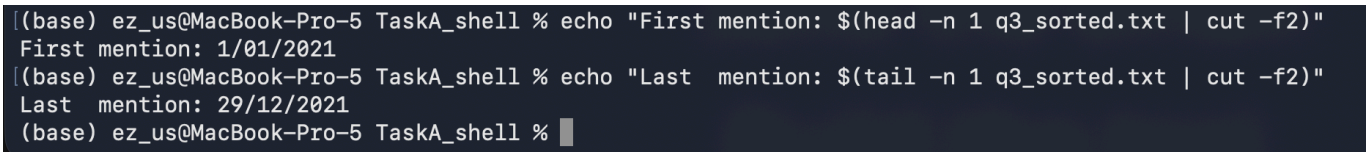
Use `awk` to keep just the rows whose column 7(`address`) contains the literal `Mount Dandenong`. Then sort date.

Code

```
# Switch the whole session to byte-mode (fixes the Illegal byte sequence)
export LC_ALL=C

# Confirm — should now print "C"
echo $LC_ALL

# Step 1: keep only rows whose address (col 7) contains "Mount Dandenong"
awk -F, '$7 ~ /Mount Dandenong/' filtered_property.csv > q3_matches.csv
```

```
wc -l q3_matches.csv                # how many transactions matched?

# Step 2: extract sold_time (col 4), drop any stray CRs
cut -d, -f4 q3_matches.csv | tr -d '\r' > q3_dates.txt

head -n 3 q3_dates.txt              # sanity: these should be DD/MM/YYYY

# Step 3: convert "D/M/YYYY" to "YYYY-MM-DD<TAB>original", sort
awk -F/ '{ printf "%04d-%02d-%02d\t%s\n", $3, $2, $1, $0 }' q3_dates.txt |
sort  > q3_sorted.txt

head -n 2 q3_sorted.txt

# Step 4: read off first and last
echo "First mention: $(head -n 1 q3_sorted.txt | cut -f2)"
echo "Last  mention: $(tail -n 1 q3_sorted.txt | cut -f2)"
```

Screenshot



Answer

- **First** (earliest) mention of "Mount Dandenong" in `address`: **1 January 2021**
- **Last** (latest) mention: **29 December 2021**

Explanation

- `awk -F, '$7 ~ /Mount Dandenong/'` column 7 contains the `Mount Dandenong` . Awk regexes are case-sensitive, so this matches the spec's requirement.
- `cut -d, -f4` extracts the `sold_time` column (now a date with no time, after Q2b). `tr -d '\r'` removes any `return`.
- Sort year-month-day.
- `echo` earliest an last line.

Q4 — First/last sold_time: odd month, Townhouse, area > 300

Approach

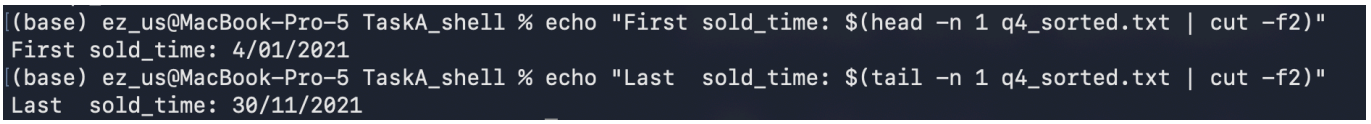
Filter `odd month, Townhouse, area > 300`, remove `NA`. Then Sort and Print.

Code

```
# Step 1: keep only Townhouse rows
```

```
awk -F, '$12 == "Townhouse"' filtered_property.csv > q4_step1_townhouse.csv
wc -l q4_step1_townhouse.csv

# Step 2: among Townhouses, keep rows where area is a plain number AND >
300.
# Patent1 of $11 matches a non-negative decimal (no NA, no "Xha", no "qq");
# Pattern 2: numeric value of $11 (forced by + 0) is greater than 300
awk -F, '$11 ~ /^[0-9]+(\.[0-9]+)?$/ && ($11 + 0) > 300'
q4_step1_townhouse.csv > q4_step2_area.csv
wc -l q4_step2_area.csv

# Step 3: keep only rows whose sold_time month is odd (1, 3, 5, 7, 9, 11)
# split($4, d, "/") gives d[1]=DD, d[2]=MM, d[3]=YYYY
# (d[2] + 0) % 2 == 1 is true only for odd months
awk -F, '{
  split($4, d, "/")
  if ((d[2] + 0) % 2 == 1) print
}' q4_step2_area.csv > q4_step3_oddmonth.csv
wc -l q4_step3_oddmonth.csv

# Step 4: extract sold_time, build sortable key, sort
cut -d, -f4 q4_step3_oddmonth.csv | tr -d '\r'  | awk -F/ '{ printf "%04d-
%02d-%02d\t%s\n", $3, $2, $1, $0 }' | sort  > q4_sorted.txt

# Step 5: report first and last
echo "First sold_time: $(head -n 1 q4_sorted.txt | cut -f2)"
echo "Last  sold_time: $(tail -n 1 q4_sorted.txt | cut -f2)"
```

Screenshot



Answer

- **First** matching `sold_time`: **4 January 2021**
- **Last** matching `sold_time`: **30 November 2021**

Explanation

- Filter (1) Townhouse (2) area > 300 (3) odd month.
- Process `NA`, and other wrong valua filter by regex `^[0-9]+(\.[0-9]+)?$`. `$11 + 0` forces numeric context so the comparison is numerical (so `"800" > 300` is true and `"50" > 300` is false; without `+ 0`, awk would compare as strings and get `"50" > "300"` as true, which is wrong).
- `split($4, d, "/")` splits `"4/01/2021"` into `d[1]=4`, `d[2]=01`, `d[3]=2021`. `(d[2] + 0) % 2 == 1` find odd month. `+ 0` force ensure data format is number. i.e. `"01"` to be treated as the number 1, not as a string.

Q5 — How many unique property types?

Approach

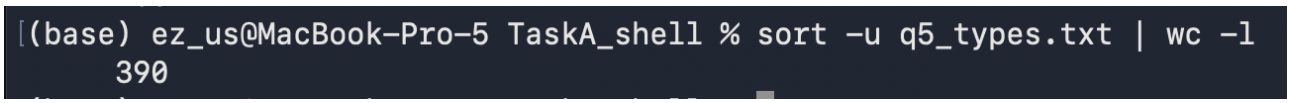
The literal answer is the number of distinct values in `property_type`, but inspection shows the column is **highly contaminated** with `area` numbers, an address fragment, and an `NA`. So we report the raw count *and* a cleaned count, with the wrangling spelled out.

Code

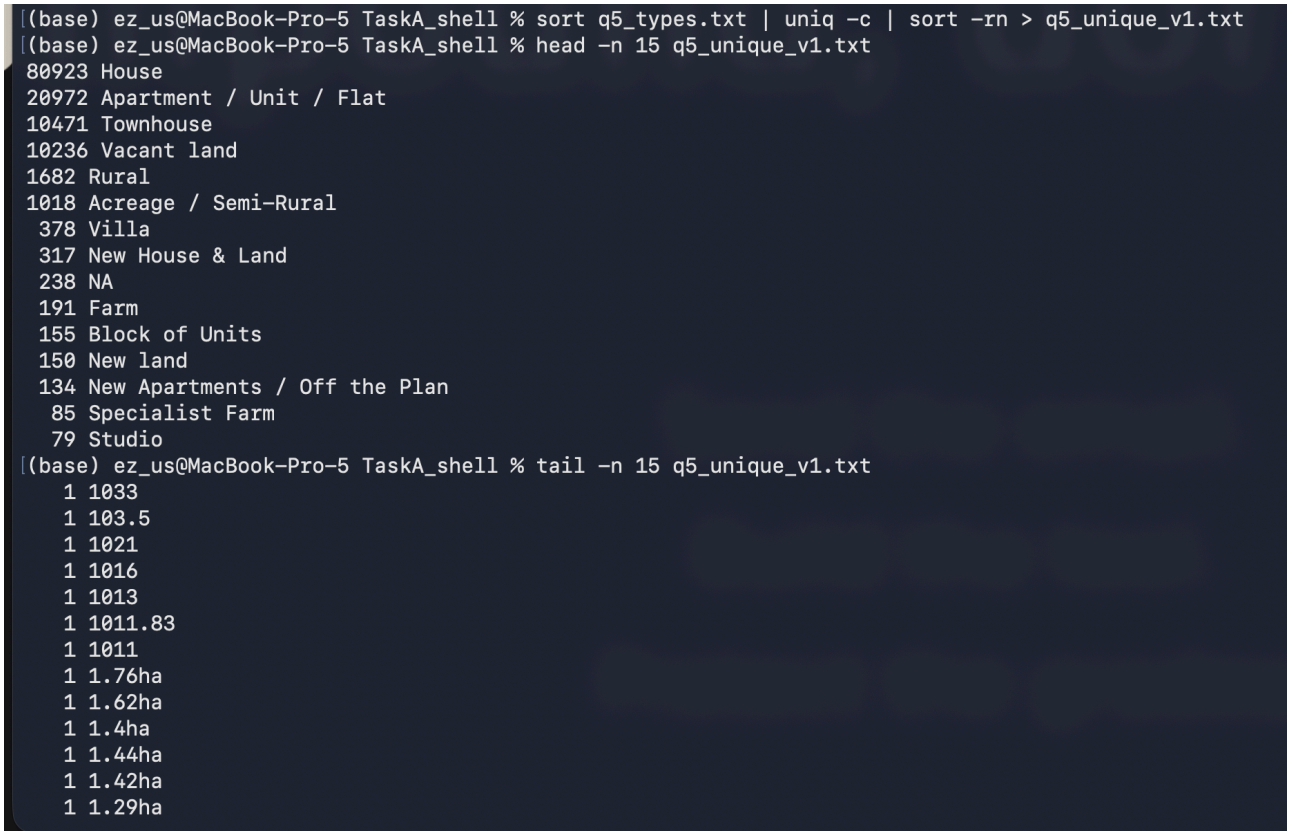
```
# Step 1: extract property_type column (col 12), skip header
tail -n +2 filtered_property.csv | cut -d, -f12 > q5_types.txt
wc -l q5_types.txt

# Step 2: raw distinct count
sort -u q5_types.txt | wc -l

# Step 3: see the counts of various type — common types first, then the
tail
sort q5_types.txt | uniq -c | sort -rn > q5_unique_v1.txt
head -n 15 q5_unique_v1.txt  # top 15 most common
tail -n 15 q5_unique_v1.txt  # bottom 15. You can see them is meanless. So
filter out them next

# Step 4: filter out garbage. Each grep -vE drops one category:
#   - "^NA$"
#   - "^[0-9]+(\.[0-9]+)?(ha)?$"  — pure numbers, optionally Xha
#   - "^[0-9]+ .* VIC "  — address-like (e.g. "20 ... VIC 3984")
sort -u q5_types.txt  | grep -vE '^NA$' | grep -vE '^[0-9]+(\.[0-9]+)?(ha)?
$' | grep -vE '^[0-9]+ .* VIC '  > q5_unique_v2.txt

wc -l q5_unique_v2.txt
cat q5_unique_v2.txt

# Step 5: case-collapsed count ("Vacant Land" and "Vacant land" merge)
tr 'A-Z' 'a-z' < q5_unique_v2.txt | sort -u | wc -l
```

Screenshot

- Step 2

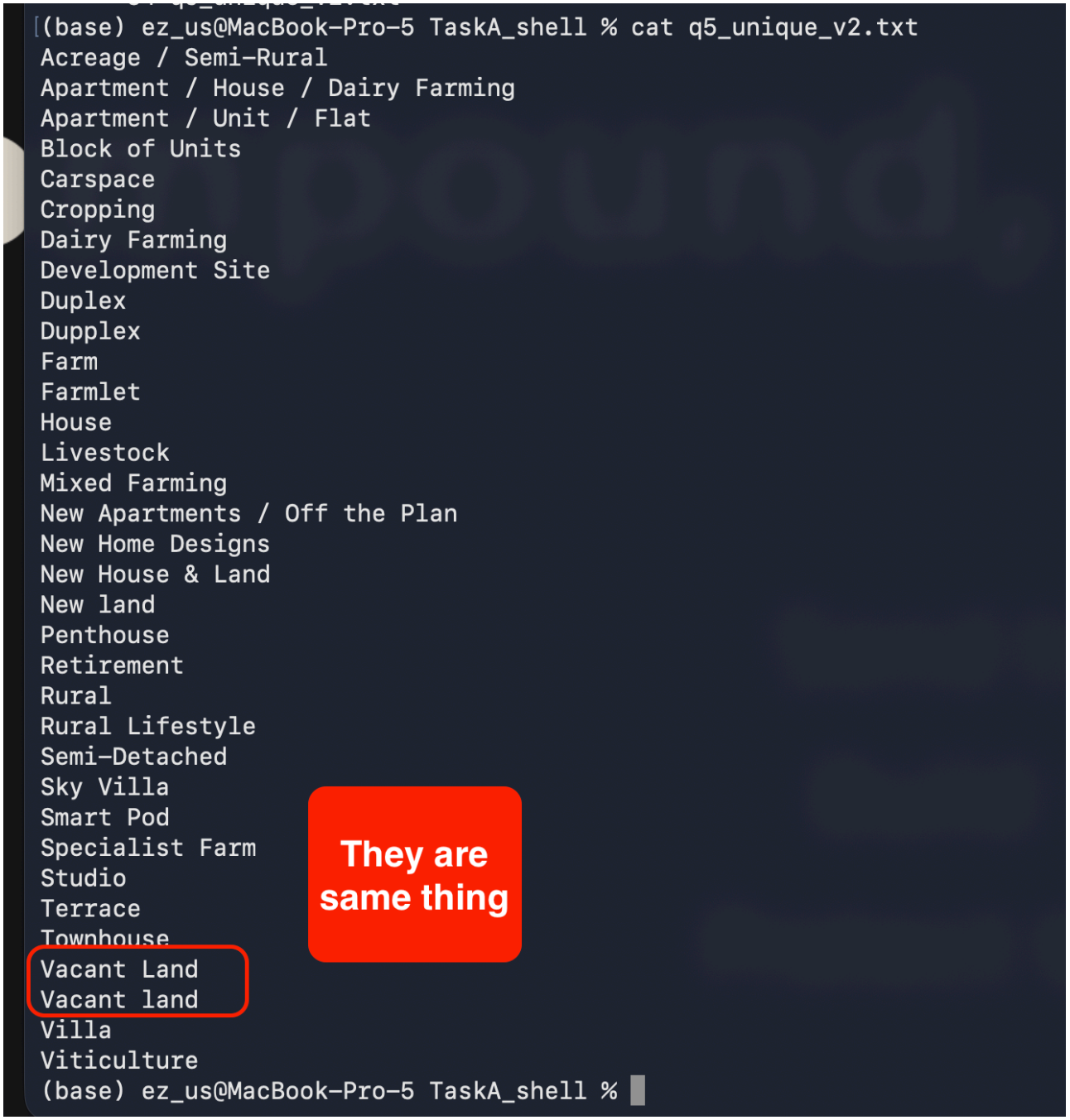


- Step 3



```
(base) ez_us@MacBook-Pro-5 TaskA_shell % sort q5_types.txt | uniq -c | sort -rn > q5_unique_v1.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell % head -n 15 q5_unique_v1.txt
 80923 House
 20972 Apartment / Unit / Flat
 10471 Townhouse
 10236 Vacant land
  1682 Rural
  1018 Acreage / Semi-Rural
   378 Villa
   317 New House & Land
   238 NA
   191 Farm
   155 Block of Units
   150 New land
   134 New Apartments / Off the Plan
    85 Specialist Farm
    79 Studio
(base) ez_us@MacBook-Pro-5 TaskA_shell % tail -n 15 q5_unique_v1.txt
     1 1033
     1 103.5
     1 1021
     1 1016
     1 1013
     1 1011.83
     1 1011
     1 1.76ha
     1 1.62ha
     1 1.4ha
     1 1.44ha
     1 1.42ha
     1 1.29ha
```

- Check result if they are purely unique or exist



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Answer

- **Raw distinct count**: 390 distinct strings in `property_type`. = Step 2
- **unique_v1 property types after removing garbage data(step 3)**: **34**.
- **After standarding case variants** (`Vacant Land` ↔ `Vacant land`): **33**.

I report **34** (or **33**) as the meaningful answer, because the other 356 raw values are not property types — 354 are `area` numbers misplaced into the column, 1 is `NA`, and 1 is an address string.

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % head -n 100 q5_unique_v1.txt
80923 House
20972 Apartment / Unit / Flat
10471 Townhouse
10236 Vacant land
1682 Rural
1018 Acreage / Semi-Rural
 378 Villa
 317 New House & Land
 238 NA
 191 Farm
 155 Block of Units
 150 New land
 134 New Apartments / Off the Plan
  85 Specialist Farm
  79 Studio
  45 Rural Lifestyle
  35 Development Site
  33 Carspace
  22 Semi-Detached
  21 Livestock
  16 Terrace
  13 Duplex
  12 4000
  10 Retirement
  10 Mixed Farming
   9 8.09ha
   9 1
   5 392
   5 343
   5 2000
   4 Farmlet
   4 Dairy Farming
   4 862
   4 796
   4 2.02ha
   4 2
   4 1022
   3 930
   3 667
   3 576
   3 509
   3 4
```

Explanation

- `-u` find the unique value, Extract to single test file. `wc -l` counts lines of the new file.
- `sort | uniq -c | sort -rn` chain — sort alphabetically so duplicates are adjacent, `uniq -c` counts each group, `sort -rn` re-sorts by count (numeric, big to small) so the most common types come first. Check the head for valid property names and the tail for meanless data(garbage).

- Filtering: Using `grep -vE` to remove the three patterns target the three garbage shapes spotted in step 3.
- Collapse: `tr 'A-Z' 'a-z'` converts(transfer) uppercase to lowercase; then `sort -u | wc -l` recounts distinct values. `Vacant Land` (1 row) and `Vacant land` (10,236 rows) merge, dropping the count from 34 to 33.

Q6 — Description column investigations

Ground setup: extract description text for searching

So extract the 14th - `description` - column first

```
# Build a description-only file used by Q6a and Q6b, and transfer clean to
lowercase format(It mentioned ignore cases in question)
tail -n +2 filtered_property.csv | cut -d, -f14 | tr 'A-Z' 'a-z' >
q6_descriptions.txt

wc -l q6_descriptions.txt  # 127706
```

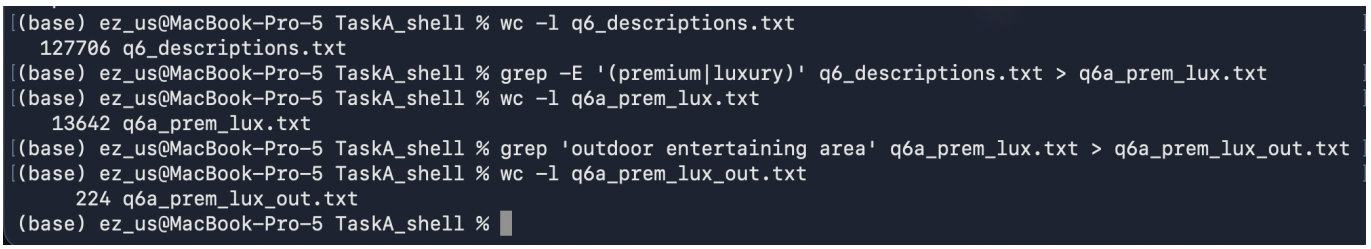
Q6a — Premium/luxury properties with outdoor entertaining areas

Code

```
# Step 1: keep descriptions premium OR luxury
grep -E '(premium|luxury)' q6_descriptions.txt > q6a_prem_lux.txt
wc -l q6a_prem_lux.txt

# Step 2: among those, keep ones mentioning "outdoor entertaining area"
grep 'outdoor entertaining area' q6a_prem_lux.txt > q6a_prem_lux_out.txt
wc -l q6a_prem_lux_out.txt
```

Screenshot



Answer

224 transaction records describe premium-or-luxury properties with an outdoor entertaining area.

Explanation

- Built `q6_descriptions.txt` once and reuse it in both Q6a and Q6b. `tr [A-Z] [a-z]` lowercase to ignore cases,
- **Step 1**: `grep -E '(premium|luxury)'` keeps lines containing either keyword. `-E` enables extended regex so the `|` means "or"
- **Step 2**: filter 'outdoor entertaining area'.

Q6b — Records mentioning a property size in their description

Approach

The spec gives two example forms: `Nm2` and `N sq metres`. I treat those as my **reference unit list**. To extend the list, I:

1. consulted standard Australian property-area units (square metre, hectare, acre) from Google,
2. inspected the description column to confirm these forms are present in the data, and to spot any forms I missed.

Code

```
# Detect "digit + unit", "1249m2" likewise string
grep -oE '[0-9]+[a-z]+[0-9]*' q6_descriptions.txt | sort | uniq -c | sort -
rn | grep -iE '(m2|sqm|acres?|hectare|ha)' | head -30

# Detect "digit + space + unit word", "758 sq metres" likewise string
grep -oE '[0-9]+ [a-z]+( [a-z]+)?' q6_descriptions.txt | sort | uniq -c |
sort -rn | grep -iE '(sq|acres?|hectares?|metres?)' | head -30

# Now I know the forms used: m2, sqm, sq metres, square metres, acres,
hectares, ha.
# Build the regex from observation:
grep -cE '[0-9]+m2|[0-9]+ ?sqm|[0-9]+ sq metres?|[0-9]+ square metres?|[0-
9]+ acres?|[0-9]+ hectares?|[0-9]+ ?ha([ .,;]|$)' q6_descriptions.txt
```

Screenshot

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % grep -oE '[0-9]+[a-z]+[0-9]*' q6_descriptions.txt | sort | uniq -c | sort
 -rn | grep -iE '(m2|sqm|acres?|hectare|ha)' | head -10
 238 650m2
 231 1000m2
 183 700m2
 173 800m2
 150 448m2
 145 400m2
 144 600m2
 140 560m2
 140 4sqm
 136 1000sqm
(base) ez_us@MacBook-Pro-5 TaskA_shell % grep -oE '[0-9]+ [a-z]+( [a-z]+)?' q6_descriptions.txt | sort | uniq -c |
 sort -rn | grep -iE '(sq|acres?|hectares?|metres?)' | head -10
 374 4 sqm
 268 5 acres
 171 5 acres of
 134 1 acre
 109 2 acres
  79 10 acres
  78 3 acres
  76 4 acre
  71 2 acres of
  70 6 acres
(base) ez_us@MacBook-Pro-5 TaskA_shell % grep -cE '[0-9]+m2|[0-9]+ ?sqm|[0-9]+ sq metres?|[0-9]+ square metres?|[0
-9]+ acres?|[0-9]+ hectares?|[0-9]+ ?ha([ .,;]|$)' q6_descriptions.txt
42066
```

Answer

42006 transaction records mention property size information in their description.

Cleanup (optional)

To remove all intermediate files but keep `filtered_property.csv`, which generate in q2b(Drop bad-ID rows and strip the time from sold_time):

```
rm q*.txt q*_step*.csv q2b_valid_ids.csv q3_matches.csv

# Snitary check
ls *.csv
```